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Executive Summary

In this homework we focus on multiple Aspen simulations in order to familiarize with the software. We use Aspen to solve these simulations by solving different types of operating data. In our first problem we used total volumetric flowrate and a temperature of waste water and different lengths for the pipes to create an Aspen simulation to calculate the minimum horsepower requirement of the pump, assuming 75% efficiency. In the second problem, we are given the template for the simulation where saturated steam is being transported in a steam pipeline. Using Aspen's different features to input the given pressure range and the lifetime roughness, we're able to use capital and operating cost equations to develop a model. We were to calculate the optimal steam pipe diameter as a function of the boiler pressure. In our third problem, we had a refrigeration with Flash Tank Economizer and Two Stages of Compression. We had to conduct a sensitivity analysis and plot the total power requirement as a function of the interstage pressure. Using the Optimization Flowsheeting option, we determined the optimal interstage pressure for the cycle. Lastly in our fourth problem, we dealt with the production of cyclohexane where we had prepared design specifications to meet the target ratios for hydrogen and liquid recycle and devised a methodology using the Aspen Plus Flowsheeting options to calculate the raw material flow rates for a production target of 100 KMTA.

Systems and Procedures

For our first problem we had to create our flowsheet using the schematic given. Looking at the schematic we were able to identify the type of equipment needed at every node and the pipe needed at every arrow in the schematic. We started off with a pump and our first two nodes were splitter and the next two nodes were mixers. At every arrow a pipe was inserted going to that specific node. Next, we inserted our component of water and inserted our givens for the stream going into the pump and the segment and length of every pipe in our flowsheet. Once the givens were inputted, initial guesses were made for the split fraction on the first splitter until the pressure out the first splitter matched the pressure out the second splitter. The second splitter pressure was matched to the second mixer. Once we got the pressures to match, we were able to find the lowest horsepower required on the pump that matches the pressure into the pump to the pressure out the second mixer at 1 atm.

For our second problem, we were given a flow sheet already prepared. The first thing we did was insert our givens in order to have the simulation run. Next, we used the sensitivity analysis where we defined our variables of diameter, density, pressure in and pressure out. In the design specs, we added diameter and pressure variables we wanted to change. We used the equation given for CAPEX and OPEX and Cost in our enhanced learning PDF to write Fortran code. We ran our simulation and were able to obtain results that gave us the most optimal diameter at the lowest cost.

The third problem also gave us a flow sheet that was prepared. We ensured the given information was written in the simulation. The chiller consisted of vapor fraction of 1 and the condenser a vapor fraction of 0. This was important to notate as we did not have to write any pressure for these systems. We ran the system with guesses for the pressure valves. Once values

were given, we created a transfer block of the first compressor's pressure to valve 1 out pressure. Additionally, a design spec around the chiller was made to meet the target of 3 MW. Using sensitivity analysis, we were able to find the COP value.

The fourth problem also gave us a flow sheet that was prepared. Using Aspen, we had to create a production target of 100 KMTA. This was done with a design spec that allowed us to meet the criteria by changing the ratios of hydrogen and liquid recycle.

Results

1. Problem A – Waste Water Treatment

- i. With the given parameters, we were able to run the simulation and were given the minimum horsepower requirement of the pump at 75% efficiency of: **521 hp**
- ii. Additional Data:
 - i. PFD – See *Figure 1* in Appendix 2
 - ii. Material Stream Data – See *Table 1* in Appendix 3

2. Problem B – Optimal Diameter of a Stream Pipe

- i. With the given data and using the article for capital and operating costs, we were able to get the optimal steam pipe diameter of different pressures (*Table 2*).
- ii. Additional Data:
 - i. PFD – See *Figure 2* in Appendix 2
 - ii. Detailed results – See *Table 2* in Appendix 3

3. Problem C – Refrigeration with Flash Tank Economizer and Two Stages of Compression

- i. With the given data, we were able to conduct a sensitivity analysis to plot the total power required along with COP of the cycle and maximum thermodynamic COP of: **COP = 2.07 and Ideal COP = 2.09, Compressor 1 = 710.9 kW and Compressor 2 = 724.9 kW, and optimal pressure = 4.97 bar**
- ii. We can conclude that the optimal pressure puts the COP close to the ideal COP.
- iii. Additional Data:
 - i. PFD – See *Figure 3* in Appendix 2
 - ii. Results Data – See *Table 3* in Appendix 3
 - iii. Sensitivity Graph – See *Figure 5* in Appendix 3

4. Problem D – Cyclohexane Production

- i. With the given data, we were able to design a specification to meet the target ratio of hydrogen and liquid recycle to prepare a production target of 100 KMTA.
- ii. Additional Data:
 - i. PFD – See *Figure 4* in Appendix 2
 - ii. Material Stream Data – See *Table 4* in Appendix 3

Conclusion

In conclusion we see how the utilization of Aspen can help solve many different types of problems. Using Aspen in problem one, we can create our PFD and run the simulation to calculate the minimum horsepower requirement of the pump. Furthermore, in problem two we used Aspen to develop a model for the calculation of the optimal steam pipe diameter as a function of the boiler pressure by using equations given for the capital and operating costs. We learned how to use Aspen for the Optimization Flowsheeting option to determine the optimal interstage pressure for the cycle in problem 3. Once we did this we were able to calculate our COP and compare it to the maximum thermodynamic COP. We got to see how we can use Aspen to create the best design specifications for a simulation knowing what our target ratios are for the cyclohexane production. This homework gave us a variety of ways to understand the different aspects Aspen as a tool can provide for process design.

Appendix 1

$$Capex = mD^nL(1 + F)(a + b)$$

$$Opex = A(\Delta P)\left(\frac{G}{\rho}\right)(m)(n)(1 + F)(a + b)$$

$$Cost = Capex + Opex$$

Table 1. Parameters for calculating the simplifying term A. Source: Ref 3.		
Parameter	Example 1	Example 2
Y	8,000 hr/yr	8,000 hr/yr
C_F	0.16 \$/Nm ³	0.29 \$/kg
J	0.5	0.5
m	330	330
n	1.5	1.5
E	0.92	0.88
K	36 MJ/ Nm ³	40 MJ/kg
F	6.5	6.5
$a+b$	0.2	0.2
A	281×10 ⁻⁶	479×10 ⁻⁶

Appendix 2

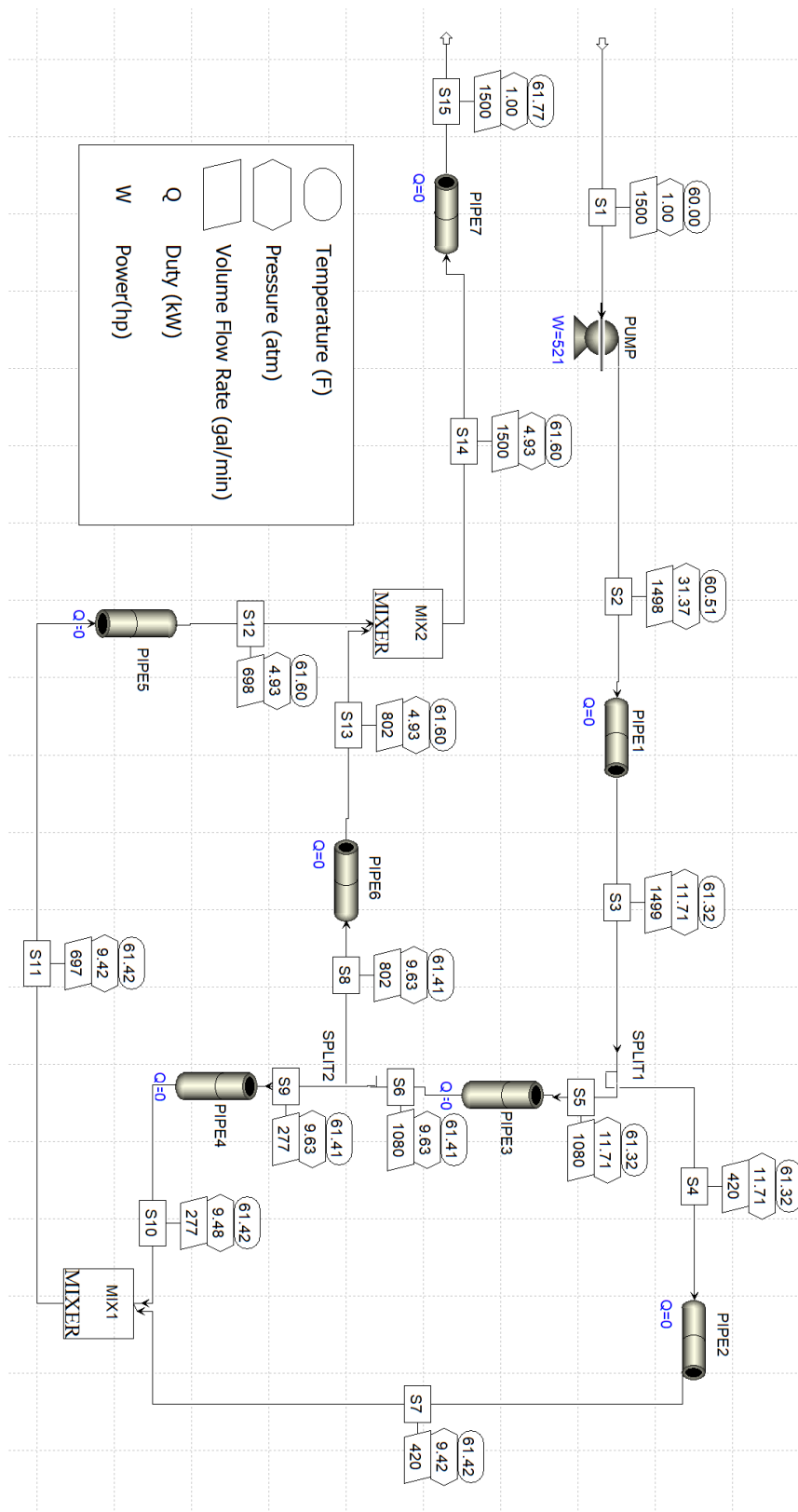


Figure 1 – Waste Water Treatment PFD (Problem A)

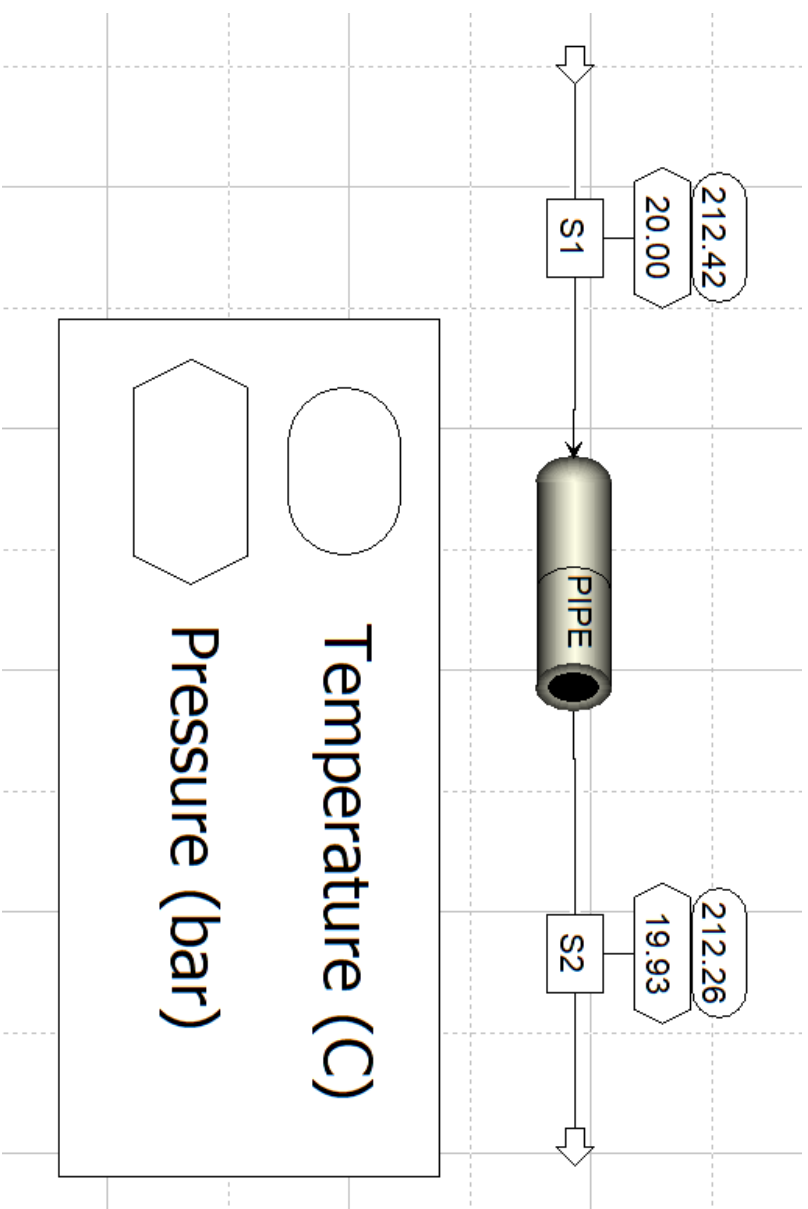


Figure 2 – Optimal Diameter of a Steam Pipe PFD (Problem B)

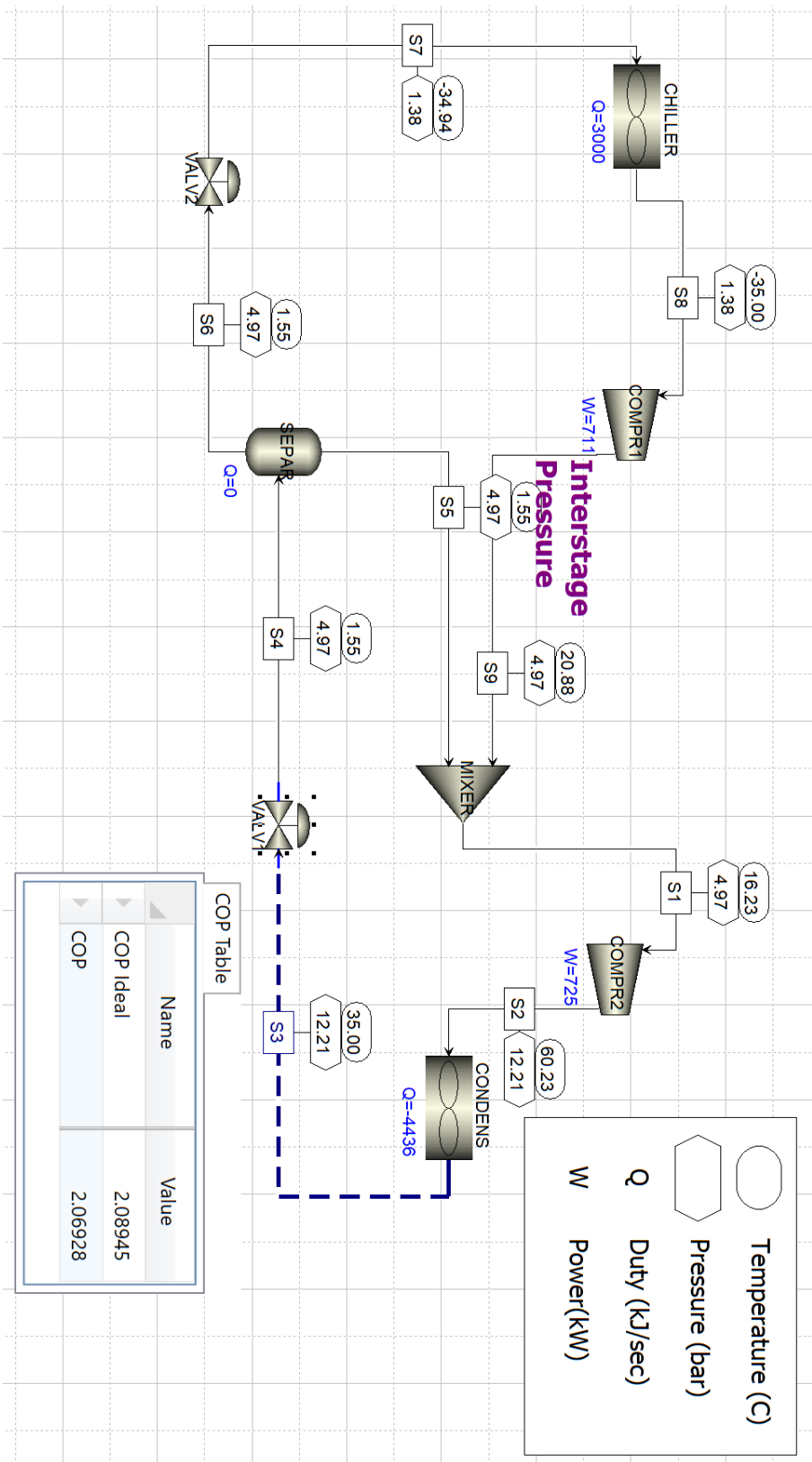


Figure 3 – Refrigeration with Flash Tank Economizer and Two Stages of Compression (Problem C)

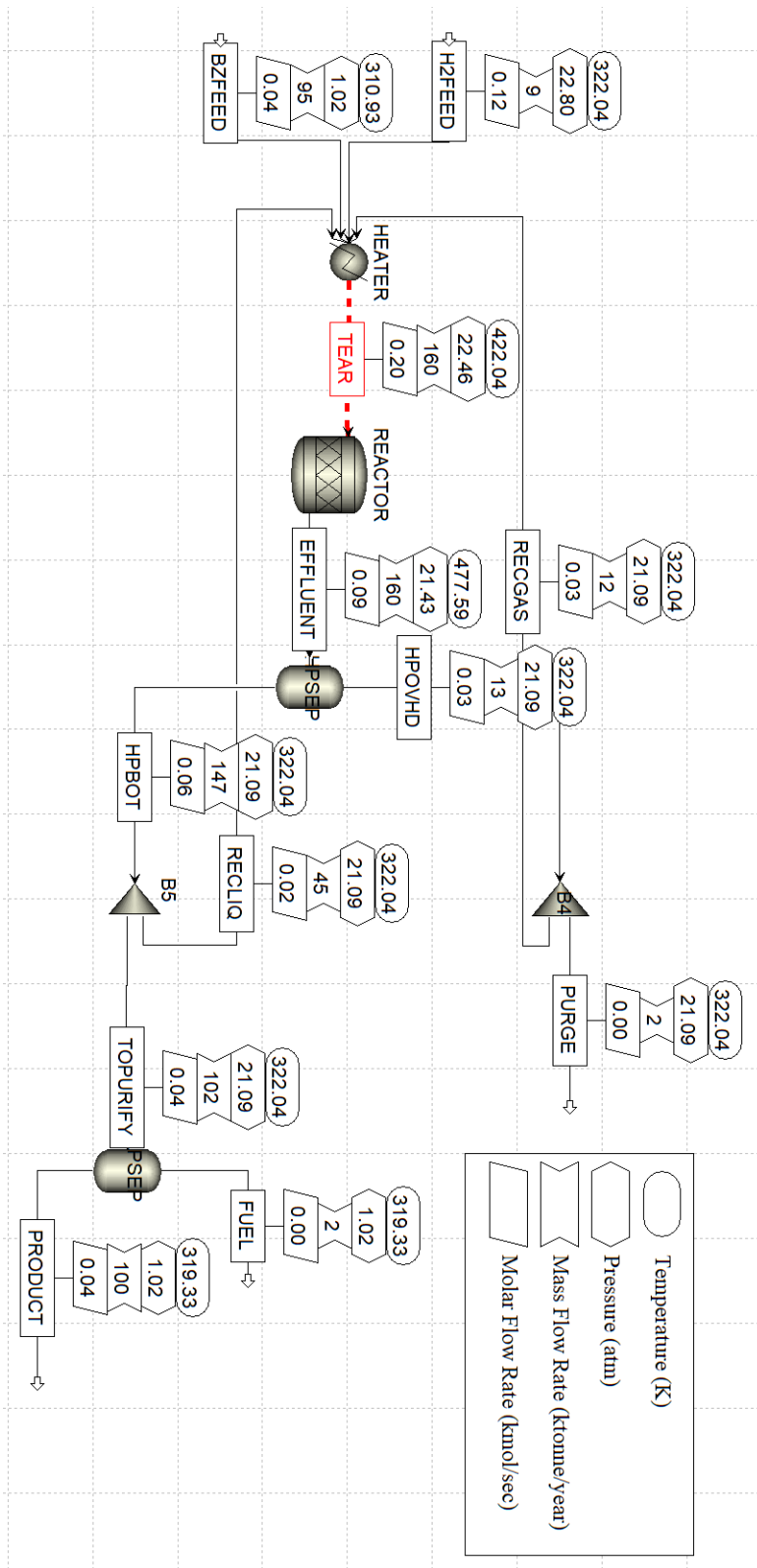


Figure 4 – Cyclohexane Production PFD (Problem D)

Material Stream Results for Problem A																		
Stream Name	Units	S1	S10	S11	S12	S13	S14	S15	S2	S3	S4	S5	S6	S7	S8	S9		
From			PIPE4	MIX1	PIPE5	PIPE6	MIX2	PIPE7	PUMP	PIPE1	SPLIT1	SPLIT1	PIPE3	PIPE2	SPLIT2	SPLIT2		
To		PUMP	MIX1	PIPE5	MIX2	MIX2	PIPE7		PIPE1	SPLIT1	PIPE2	PIPE3	SPLIT2	MIX1	PIPE6	PIPE4		
MIXED Substream																		
Phase		Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid		
Temperature	F	60	61.4153	61.4175	61.604	61.6038	61.604	61.7672	60.5058	61.3224	61.3224	61.3224	61.4088	61.4175	61.4088	61.4088		
Pressure	atm	1	9.47523	9.42381	4.93022	4.93408	4.93022	0.997512	31.3699	11.7139	11.7139	11.7139	9.6335	9.42381	9.6335	9.6335		
Molar Vapor Fraction		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Molar Liquid Fraction		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
Molar Enthalpy	J/kmol	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08	-2.87E+08		
Mass Enthalpy	J/kg	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07	-1.59E+07		
Molar Entropy	J/kmol-K	-165575	-165372	-165372	-165344	-165344	-165344	-165319	-165511	-165387	-165387	-165387	-165373	-165372	-165373	-165373		
Mass Entropy	J/kg-K	-9190.8	-9179.57	-9179.55	-9177.98	-9177.98	-9177.98	-9176.6	-9187.24	-9180.35	-9180.35	-9180.35	-9179.62	-9179.55	-9179.62	-9179.62		
Molar Density	kmol/cum	55.4557	55.4707	55.4705	55.4578	55.4579	55.4578	55.4467	55.5323	55.477	55.477	55.477	55.4711	55.4705	55.4711	55.4711		
Mass Density	kg/cum	999.05	999.32	999.317	999.089	999.089	999.089	998.888	1000.43	999.433	999.433	999.433	999.328	999.317	999.328	999.328		
Enthalpy Flow	kW	-1.50E+06	-278355	-699559	-699559	-804740	-1.50E+06	-1.50E+06	-1.50E+06	-1.50E+06	-421204	-1.08E+06	-1.08E+06	-421204	-804740	-278355		
Average MW		18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153	18.0153		
Mole Flows	kmol/hr	18893	3495.97	8786.02	8786.02	10107	18893	18893	18893	18893	5290.05	13603	13603	5290.05	10107	3495.97		
Mass Flows	gm/min	5.67E+06	1.05E+06	2.64E+06	2.64E+06	3.03E+06	5.67E+06	5.67E+06	5.67E+06	5.67E+06	1.59E+06	4.08E+06	4.08E+06	1.59E+06	3.03E+06	1.05E+06		
Volume Flow	gal/min	1500	277.485	697.374	697.533	802.409	1499.94	1500.24	1497.93	1499.43	419.839	1079.59	1079.7	419.888	802.217	277.483		

Table 1 – Material Stream Data (Problem A)

Appendix 3

Boiler Pressure, bar	Optimal Diameter, mm (from paper)	ASPEN Plus Optimal Diameter (mm)	CAPEX (\$/yr)	OPEX(\$/yr)	TOTAL COST (\$/yr)
1	122.6	122.9	\$ 21.33	\$ 12,901.20	\$12,922.53
2	103.2	103.5	\$ 16.48	\$ 13,490.50	\$13,506.98
5	79.6	79.8	\$ 11.16	\$ 14,271.60	\$14,282.76
8	69.6	69.7	\$ 9.11	\$ 14,642.10	\$14,651.21
10	65.3	65.2	\$ 8.24	\$ 14,801.00	\$14,809.24
15	58.1	57.9	\$ 6.74	\$ 15,044.50	\$15,051.24
20	53.4	53.3	\$ 6.11	\$ 15,166.00	\$15,172.11

Table 2 – Steam Pipe Optimal Diameter on Different Pressures (Problem B)

Material Stream Results for Problem C										
Stream Name	Units	S1	S2	S3	S4	S5	S6	S7	S8	S9
Description										
From		MIXER	COMPR2	CONDENS	VALV1	SEPAR	SEPAR	VALV2	CHILLER	COMPR1
To		COMPR2	CONDENS	VALV1	SEPAR	MIXER	VALV2	CHILLER	COMPR1	MIXER
MIXED Substream										
Phase		Vapor	Vapor	Liquid	Mixed	Vapor	Liquid	Mixed	Vapor	Vapor
Temperature	C	6.46596	60.272	35	-5.44167	-5.44167	-5.44167	-34.9391	-35	11.1037
Pressure	bar	4	12.21	12.2062	4	4	4	1.38	1.37656	4
Molar Vapor Fraction		1	1	0	0.283155	1	0	0.167311	1	1
Molar Liquid Fraction		0	0	1	0.716845	0	1	0.832689	0	0
Molar Enthalpy	kJ/kmol	-106535	-103279	-119542	-119542	-107403	-124337	-124337	-109030	-106192
Mass Enthalpy	kJ/kg	-2415.95	-2342.12	-2710.92	-2710.92	-2435.63	-2819.66	-2819.66	-2472.53	-2408.18
Molar Entropy	kJ/kmol-K	-286.525	-284.049	-336.543	-335.042	-289.695	-352.953	-352.127	-287.848	-285.309
Mass Entropy	kJ/kg-K	-6.49767	-6.44153	-7.63197	-7.59791	-6.56957	-8.00411	-7.98537	-6.52768	-6.4701
Molar Density	mol/cc	0.000187601	0.000526362	0.0107935	0.000672781	0.000198367	0.0121621	0.000423065	7.26E-05	0.000183765
Mass Density	gm/cc	0.00827257	0.0232107	0.475955	0.0296673	0.00874731	0.536308	0.0186557	0.00320078	0.00810339
Enthalpy Flow	kJ/sec	-29126.8	-28236.7	-32683	-32683	-8314.58	-24368.4	-24368.4	-21368.4	-20812.3
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965
Mole Flows	kmol/hr	984.246	984.246	984.246	984.246	278.694	705.552	705.552	705.552	705.552
Mass Flows	kg/hr	43401.8	43401.8	43401.8	43401.8	12289.4	31112.4	31112.4	31112.4	31112.4
Volume Flow	cum/hr	5246.48	1869.9	91.1889	1462.95	1404.94	58.0121	1667.71	9720.26	3839.43

Table 3 - Refrigeration with Flash Tank Economizer and Two Stages of Compression (Problem C)

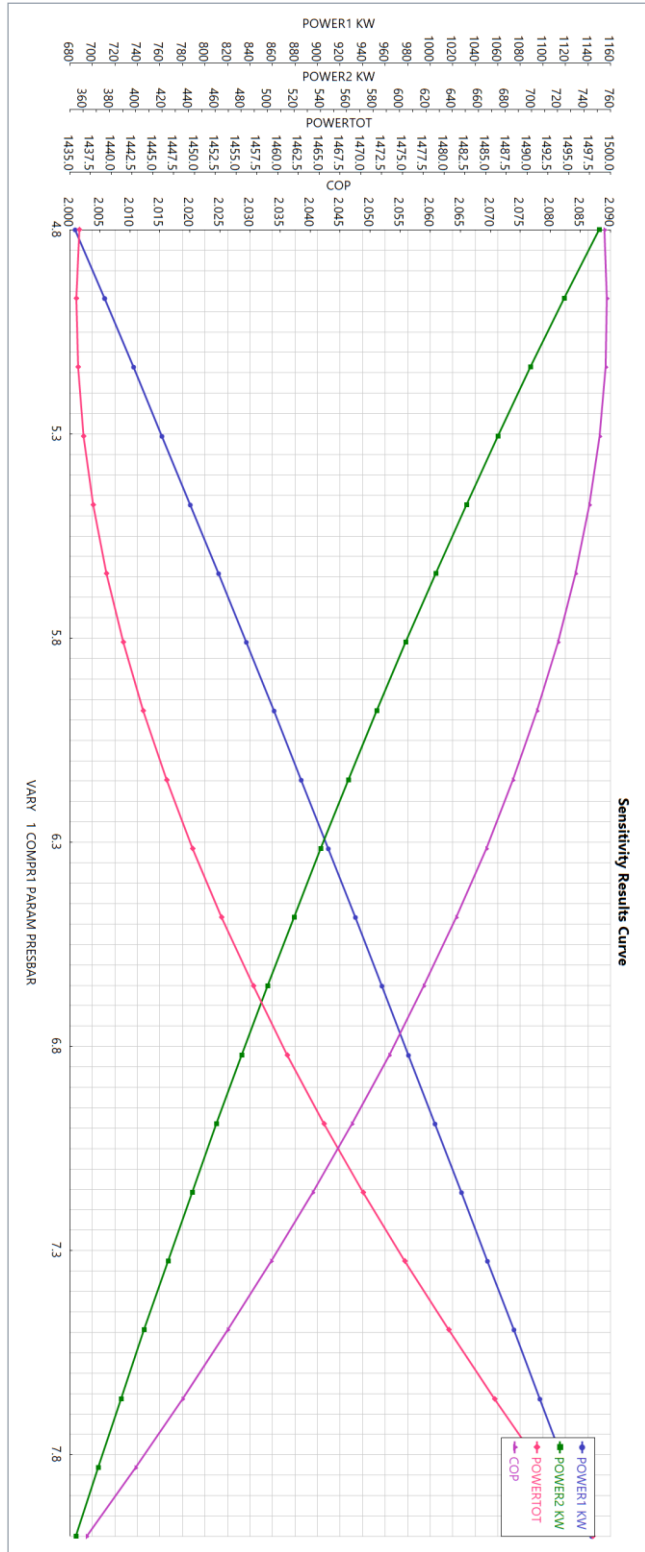


Figure 5 – Sensitivity Results Curve (Problem C)

Material Stream Results for Problem D													
Stream Name	Units	BZFEED	EFFLUENT	FUEL	H2FEED	HPBOT	HPVOHD	PRODUCT	PURGE	REGAS	RECLIQ	TEAR	TOPURIFY
Description													
From			REACTOR	LPSEP		HPSEP	HPSEP	LPSEP	B4	B4	B5	HEATER	B5
To		HEATER	HPSEP		HEATER	B5	B4			HEATER	HEATER	REACTOR	LPSEP
MIXED Substream													
Phase		Liquid	Vapor	Vapor	Vapor	Liquid	Vapor	Liquid	Vapor	Vapor	Liquid	Vapor	Liquid
Temperature	K	310.928	477.594	319.327	322.039	322.039	322.039	319.327	322.039	322.039	322.039	422.039	322.039
Pressure	atm	1.02069	21.4345	1.02069	22.7954	21.0942	21.0942	1.02069	21.0942	21.0942	21.0942	22.4552	21.0942
Molar Vapor Fraction		0	1	1	1	0	1	0	1	1	0	1	0
Molar Liquid Fraction		1	0	0	0	1	0	1	0	0	1	0	1
Molar Enthalpy	J/kmol	5.09E+07	-7.56E+07	-7.91E+07	-777549	-1.49E+08	-3.88E+07	-1.53E+08	-3.88E+07	-3.88E+07	-1.49E+08	5.46E+06	-1.49E+08
Mass Enthalpy	J/kg	652179	-1.30E+06	-2.18E+06	-320453	-1.82E+06	-2.79E+06	-1.82E+06	-2.79E+06	-2.79E+06	-1.82E+06	218137	-1.82E+06
Molar Entropy	J/kmol-K	-246579	-316796	-190149	-24322.7	-581726	-64490.6	-601435	-64490.6	-64490.6	-581726	-79500.5	-581726
Mass Entropy	J/kg-K	-3156.67	-5463.67	-5245.22	-10024.2	-7132.24	-4633.97	-7158.48	-4633.97	-4633.97	-7132.24	-3175.65	-7132.24
Molar Density	kmol/cum	10.9994	0.626464	0.0392335	0.851464	9.06858	0.799699	8.94504	0.799699	0.799699	9.06858	0.657566	9.06858
Mass Density	kg/cum	859.204	36.3238	1.42229	2.066	739.66	11.1293	751.537	11.1293	11.1293	739.66	16.4618	739.66
Enthalpy Flow	Watt	1.96E+06	-6.62E+06	-161830	-93633.8	-8.49E+06	-1.18E+06	-5.76E+06	-144167	-1.04E+06	-2.57E+06	1.11E+06	-5.92E+06
Average MW		78.1136	57.9823	36.2518	2.42641	81.5629	13.9169	84.0172	13.9169	13.9169	81.5629	25.0344	81.5629
Mole Flows	kmol/sec	0.0384734	0.087661	0.0020467	0.120422	0.0571034	0.0305576	0.0377482	0.00371835	0.0268392	0.0173106	0.203032	0.0397928
Mass Flows	ktonne/year	94.84	160.401	2.33915	9.22088	146.98	13.4204	100.085	1.63304	11.7874	44.5563	160.401	102.424
Volume Flow	cum/sec	0.00349777	0.13993	0.0521155	0.141429	0.00629685	0.0382113	0.00422001	0.00464968	0.0335617	0.00190886	0.308763	0.00438799

Table 4 – Cyclohexane Production Material Streams (Problem D)

References

Genic, Srbislav. "A Shortcut to Determine Optimal Steam Pipe Diameter." *AIChE*, 2 Aug. 2018, www.aiche.org/resources/publications/cep/2018/august/shortcut-determine-optimal-steam-pipe-diameter.